

GENERALIZATION ABILITY OF NEURAL NETWORKS AND OTHER COMBINED CLASSIFIERS

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1. BACKGROUND

A fundamental goal in machine learning is to build models that work not just on the data they’ve seen, but on data they haven’t seen. This is the problem of *generalization*.

Consider building a cat detector: we’re given sample images $X_1, \dots, X_n$ with labels $Y_i = \pm 1$ indicating whether each image contains a cat. Using this training data, we learn a classifier f that maps images to predictions via $\text{sgn}(f(x))$. The true test of success is the generalization error:

$$(1) \quad \mathcal{E}(f) := \mathbb{P}(\text{sgn}(f(X)) \neq Y) = \mathbb{P}(Yf(X) \leq 0).$$

This measures the probability our classifier fails on a new image from the same distribution. The fundamental challenge is: we want $\mathcal{E}(f)$ to be small, but we can only observe performance on our training data through the training error:

$$(2) \quad \mathcal{E}_n(f) := \frac{1}{n} \sum_{i=1}^n \mathbb{I}(Y_i f(X_i) \leq 0).$$

If we fixed f in advance, the training error would reliably estimate $\mathcal{E}(f)$ (via the SLLN). But in practice, we choose f to minimize the training error, creating a subtle dependency where the same data is used for both selection and evaluation. This means we need something stronger: we need the training error to approximate the generalization error well for all classifiers f in our model class $\mathcal{F}$ simultaneously. That is, we need to bound the generalization gap:

$$(3) \quad Z := \sup_{f \in \mathcal{F}} (\mathcal{E}(f) - \mathcal{E}_n(f)).$$

Indeed, for any $f \in \mathcal{F}$, we can write:

$$\mathcal{E}(f) = \mathcal{E}_n(f) + (\mathcal{E}(f) - \mathcal{E}_n(f)) \leq \mathcal{E}_n(f) + Z,$$

so controlling Z ensures good generalization for every classifier our learning algorithm might produce.

The classical solution, pioneered by Vapnik and Chervonenkis [VC71; Vap82], bounds this uniform approximation error using the VC dimension of $\mathcal{F}$. If $\mathcal{F}$ is sufficiently “simple”, good training performance implies good generalization. But herein lies the puzzle: modern models like deep neural networks use enormously complex classes $\mathcal{F}$ with huge VC dimensions [BMM99]. According to classical theory, these models should overfit terribly, yet in practice they generalize remarkably well. In this talk, we explore how Koltchinskii and Panchenko [KP02] resolved this paradox by shifting focus from whether a classifier is correct to how confidently it’s correct. Building on insights from Bartlett, Schapire, Freund, and others [Bar98; Sch+98], their key idea is that margins (not just errors) hold the secret to understanding generalization in complex models.

2. BOUNDING THE GENERALIZATION GAP

As mentioned in Section 1, we are interested in bounding the quantity Z defined by (3), that is, the worst-case generalization gap $\mathcal{E}(f) - \mathcal{E}_n(f)$ as f ranges over the function class $\mathcal{F}$. As a first step in this endeavour, we argue that with high probability Z is concentrated around its mean, allowing us to reduce the problem to estimating the deterministic quantity $\mathbb{E}Z$.

Indeed, observe that via the training error term, Z is a function of the i.i.d. sample-label pairs $\{(X_i, Y_i)\}_{i=1}^n$. As such a function, we claim that it is furthermore “stable” in the sense that replacing one of the training samples (X_i, Y_i) with an independent copy (X'_i, Y'_i) thereof changes the output by at most $1/n$. To see this, let us denote by $D = \{(X_i, Y_i)\}_{i=1}^n$ our original dataset, and by D' the same dataset with (X'_i, Y'_i) in place of (X_i, Y_i) . Moreover, let us denote by $\mathcal{E}_n(f; D)$ and $Z(D)$ (resp. $\mathcal{E}_n(f; D')$ and $Z(D')$) the corresponding values of $\mathcal{E}_n(f)$ and Z . The key observation is that for every $f \in \mathcal{F}$ we have that

$$\begin{aligned} |(\mathcal{E}(f) - \mathcal{E}_n(f; D)) - (\mathcal{E}(f) - \mathcal{E}_n(f; D'))| &= |\mathcal{E}_n(f; D) - \mathcal{E}_n(f; D')| \\ &= \frac{1}{n} |\mathbb{I}(Y_i f(X_i) \leq 0) - \mathbb{I}(Y'_i f(X'_i) \leq 0)| \leq \frac{1}{n}. \end{aligned}$$

With this in mind, we note that for any $\varepsilon > 0$ we can (by definition of the supremum) find $f_* \in \mathcal{F}$ such that $Z(D) = (\mathcal{E}(f_*) - \mathcal{E}_n(f_*; D)) + \varepsilon$, whence

$$\begin{aligned} Z(D) - Z(D') &= [(\mathcal{E}(f_*) - \mathcal{E}_n(f_*; D)) + \varepsilon] - \sup_{f \in \mathcal{F}} (\mathcal{E}(f) - \mathcal{E}_n(f; D')) \\ &\leq (\mathcal{E}(f_*) - \mathcal{E}_n(f_*; D)) - (\mathcal{E}(f_*) - \mathcal{E}_n(f_*; D')) + \varepsilon \\ &\leq \frac{1}{n} + \varepsilon. \end{aligned}$$

Taking $\varepsilon \rightarrow 0$, it follows that $Z(D) - Z(D') \leq 1/n$. By a symmetric argument with the roles of D and D' reversed, we deduce that $|Z(D) - Z(D')| \leq 1/n$, as claimed. Having established this, we can now apply the following classical concentration result to the random variable Z :

Lemma (Azuma’s Inequality). *Suppose that $g : \mathbb{R}^n \rightarrow \mathbb{R}$ is a function for which there exists positive constants $c_1, \dots, c_n > 0$ such that*

$$|g(x_1, \dots, x_i, \dots, x_n) - g(x_1, \dots, x'_i, \dots, x_n)| \leq c_i$$

for all $x_1, \dots, x_i, x'_i, \dots, x_n \in \mathbb{R}$. If $X_1, \dots, X_n$ are independent random variables, then

$$\mathbb{P}(g(X_1, \dots, X_n) - \mathbb{E}g(X_1, \dots, X_n) \geq t) \leq \exp\left(-\frac{2t^2}{\sum_{i=1}^n c_i^2}\right)$$

for every $t > 0$.

In our case all the c_i are given by $1/n$, so the bound reads as

$$\mathbb{P}(Z - \mathbb{E}Z \geq t) \leq \exp(-2nt^2)$$

for every $t > 0$. In particular, by replacing t with $t/\sqrt{n}$, we can rephrase this as the following holding with probability at least $1 - e^{-2t^2}$:

$$(4) \quad Z = \mathbb{E}Z + (Z - \mathbb{E}Z) \leq \mathbb{E}Z + \frac{t}{\sqrt{n}}.$$

Note that this is an exceptionally strong bound: the error probability decays like e^{-t^2} so that even for relatively small values of t the above holds with extremely high probability (for instance, if $t = 2$ then it holds with probability $p \approx 0.9997$). Accordingly, to estimate Z (with high probability) it suffices to estimate the deterministic quantity $\mathbb{E}Z$. To do so, we employ a symmetrization trick. Indeed, notice that if we introduce an independent “ghost copy” $\{(X'_i, Y'_i)\}_{i=1}^n$ of our dataset, then we can write

$$\mathbb{E}Z = \mathbb{E}_X \left[\sup_{f \in \mathcal{F}} (\mathbb{E}_{X'} [\mathcal{E}'_n(f)] - \mathcal{E}_n(f)) \right],$$

where $\mathcal{E}'_n(f) := \frac{1}{n} \sum_{i=1}^n \mathbb{I}(Y'_i f(X'_i) \leq 0)$ is the empirical error over this copy, and the notation $\mathbb{E}_X$ (resp. $\mathbb{E}_{X'}$) denotes expectation over just the X_i (resp. just the X'_i). Since $\mathcal{E}_n(f)$ is independent of the X'_i ,

we can then write $\mathcal{E}_n(f) = \mathbb{E}_{X'}[\mathcal{E}_n(f)]$ and apply Jensen's inequality to commute the sup with the expectation $\mathbb{E}_{X'}$ to get

$$\mathbb{E}Z \leq \mathbb{E}_{X, X'} \left[\sup_{f \in \mathcal{F}} (\mathcal{E}'_n(f) - \mathcal{E}_n(f)) \right],$$

where the notation $\mathbb{E}_{X, X'}$ denotes the double expectation $\mathbb{E}_X \mathbb{E}_{X'}$ (i.e. expectation over the product space $\Omega \times \Omega$). Let us now take a moment to look at the quantity

$$F(X_1, \dots, X_n, X'_1, \dots, X'_n) := \sup_{f \in \mathcal{F}} (\mathcal{E}'_n(f) - \mathcal{E}_n(f)) = \sup_{f \in \mathcal{F}} \frac{1}{n} \sum_{i=1}^n (\mathbb{I}(Y'_i f(X'_i) \leq 0) - \mathbb{I}(Y_i f(X_i) \leq 0)).$$

Notice that this is a measurable function of the X_i and X'_i . Because the X_i and X'_i are i.i.d., the distribution of the above is subsequently invariant under permutations of its $2n$ arguments. Since adding a minus sign in front of any one of the summands has the effect of exchanging X_i and X'_i , it follows that

$$\mathbb{E}_{X, X'} \left[\sup_{f \in \mathcal{F}} \frac{1}{n} \sum_{i=1}^n \varepsilon_i (\mathbb{I}(Y'_i f(X'_i) \leq 0) - \mathbb{I}(Y_i f(X_i) \leq 0)) \right]$$

is the same for every n -tuple $(\varepsilon_1, \dots, \varepsilon_n) \in \{-1, 1\}^n$. Accordingly, if $\varepsilon_1, \dots, \varepsilon_n$ are taken to be i.i.d. Rademacher random variables that are independent of the X_i and X'_i , then we can re-write the above as

$$\mathbb{E}_{X, X', \varepsilon} \left[\sup_{f \in \mathcal{F}} \frac{1}{n} \sum_{i=1}^n \varepsilon_i (\mathbb{I}(Y'_i f(X'_i) \leq 0) - \mathbb{I}(Y_i f(X_i) \leq 0)) \right],$$

at which point we can apply the triangle inequality to deduce

$$\begin{aligned} \mathbb{E}Z &\leq \mathbb{E}_{X, X', \varepsilon} \left[\sup_{f \in \mathcal{F}} \frac{1}{n} \sum_{i=1}^n \varepsilon_i \mathbb{I}(Y'_i f(X'_i) \leq 0) + \sup_{f \in \mathcal{F}} \frac{1}{n} \sum_{i=1}^n -\varepsilon_i \mathbb{I}(Y_i f(X_i) \leq 0) \right] \\ &= 2\mathbb{E} \left[\sup_{f \in \mathcal{F}} \frac{1}{n} \sum_{i=1}^n \varepsilon_i \mathbb{I}(Y_i f(X_i) \leq 0) \right], \end{aligned}$$

where in the final line we have used the abbreviation $\mathbb{E} = \mathbb{E}_{X, \varepsilon}$. The right-hand side here has a special name. Namely, for a class $\mathcal{H}$ of measurable functions $S \rightarrow \mathbb{R}$, we denote by $\mathcal{R}_n(\mathcal{H})$ the quantity

$$R_n(\mathcal{H}) := \mathbb{E} \left[\sup_{h \in \mathcal{H}} \frac{1}{n} \sum_{i=1}^n \varepsilon_i h(X_i) \right]$$

and call the mapping $n \mapsto R_n(\mathcal{H})$ the (one-sided) *Rademacher complexity function* of the class $\mathcal{H}$.

Remark. To motivate $R_n(\mathcal{H})$ as a notion of complexity, we note that it (roughly speaking) measures how well the function class $\mathcal{H}$ can fit random noise. Indeed, notice that

$$\sum_{i=1}^n \varepsilon_i h(X_i) = \varepsilon \cdot h(\mathbf{X}),$$

where $\varepsilon = (\varepsilon_1, \dots, \varepsilon_n) \in \{-1, 1\}^n$ is random noise and $h(\mathbf{X}) = (h(X_1), \dots, h(X_n))$ is the “prediction” vector. The dot product $\varepsilon \cdot h(\mathbf{X})$ is large when the two vectors are aligned in the sense of ε_i and $h(X_i)$ having the same sign. Thus, if the supremum of this quantity over all $h \in \mathcal{H}$ is large, then our function class is large enough (or “complex” enough) to contain an h that correlates well with the noise vector ε in this sense.

With this notation, our bound reads as

$$(5) \quad \mathbb{E}Z \leq 2R_n(\mathcal{L} \circ \tilde{\mathcal{F}}),$$

where $\mathcal{L}(u) = \mathbf{1}(u \leq 0)$ denotes our 0-1 “loss function”, $\tilde{\mathcal{F}}$ the function class $\{\tilde{f}(x, y) = yf(x) : f \in \mathcal{F}\}$, and $\mathcal{L} \circ \tilde{\mathcal{F}}$ the composed function class $\{\mathcal{L} \circ \tilde{f} : \tilde{f} \in \tilde{\mathcal{F}}\}$. Combining (4) and (5), we have established that for every $f \in \mathcal{F}$, the following holds with probability at least $1 - e^{-2t^2}$:

$$\mathcal{E}(f) \leq \mathcal{E}_n(f) + 2R_n(\mathcal{L} \circ \tilde{\mathcal{F}}) + \frac{t}{\sqrt{n}}.$$

This is a nice bound, but there are two aspects of it that would be nice to improve upon. Firstly, Rademacher random variables have less powerful theoretical properties than Gaussian random variables (e.g. in the Gaussian case we have access to Slepian’s lemma, nice tail bounds, etc). It would be nice if we

could somehow replace the Rademacher complexity term with the corresponding (one-sided) *Gaussian complexity* term, defined for a function class $\mathcal{H}$ as

$$\mathcal{G}_n(\mathcal{H}) := \mathbb{E} \left[\sup_{h \in \mathcal{H}} \frac{1}{n} \sum_{i=1}^n g_i h(X_i) \right]$$

for i.i.d. standard Gaussians $g_1, \dots, g_n$ that are independent of the X_i , and with the expectation being taken over both the g_i and the X_i (i.e. $\mathbb{E} = \mathbb{E}_{X,g}$ here). Secondly, the 0-1 loss function $\mathcal{L}$ is non-Lipschitz, meaning that the composition $\mathcal{L} \circ \tilde{\mathcal{F}}$ can obfuscate the underlying “structure” of $\tilde{\mathcal{F}}$: even small changes in $\tilde{f}$ can result in “large” changes in $\mathcal{L} \circ \tilde{f}$. In other words, we don’t really know much about $\mathcal{L} \circ \tilde{\mathcal{F}}$, making it harder to work with directly. It would be nice if we could somehow get a bound purely in terms of the function class $\tilde{\mathcal{F}}$ (and eventually $\mathcal{F}$).

The first of these goals (i.e. the transition from Rademacher complexity to Gaussian complexity) is actually quite easy to achieve. Indeed, we claim that for any function class $\mathcal{H}$, it holds that

$$R_n(\mathcal{H}) \leq \sqrt{\frac{\pi}{2}} G_n(\mathcal{H}).$$

To see this, note that we have $\mathbb{E}_g |g_i| = \sqrt{2/\pi}$, so we can write

$$\begin{aligned} R_n(\mathcal{H}) &= \mathbb{E}_{X,\varepsilon} \left[\sup_{h \in \mathcal{H}} \frac{1}{n} \sum_{i=1}^n \varepsilon_i h(X_i) \right] \\ &= \sqrt{\frac{\pi}{2}} \mathbb{E}_{X,\varepsilon} \left[\sup_{h \in \mathcal{H}} \frac{1}{n} \sum_{i=1}^n \varepsilon_i \mathbb{E}_g |g_i| h(X_i) \right] \\ &\leq \sqrt{\frac{\pi}{2}} \mathbb{E}_X \mathbb{E}_\varepsilon \mathbb{E}_g \left[\sup_{h \in \mathcal{H}} \frac{1}{n} \sum_{i=1}^n \varepsilon_i |g_i| h(X_i) \right] \\ &= \sqrt{\frac{\pi}{2}} \mathbb{E}_X \mathbb{E}_{\tilde{g}} \left[\sup_{h \in \mathcal{H}} \frac{1}{n} \sum_{i=1}^n \tilde{g}_i h(X_i) \right] = \sqrt{\frac{\pi}{2}} G_n(\mathcal{H}). \end{aligned}$$

Notice that the second last equality follows because, on the product space $\Omega_\varepsilon \times \Omega_g$, the random variables $\tilde{g}_i = \varepsilon_i |g_i|$ are independent standard Gaussians. With this, our bound improves to:

$$\mathcal{E}(f) \leq \mathcal{E}_n(f) + \sqrt{2\pi} G_n(\mathcal{L} \circ \tilde{\mathcal{F}}) + \frac{t}{\sqrt{n}} \quad \text{with probability at least } 1 - e^{-2t^2}, \text{ for every } f \in \mathcal{F}.$$

Now what we wish to do is eliminate $\mathcal{L}$ and get a bound purely in terms of $G_n(\tilde{\mathcal{F}})$. This is more difficult, since (as previously mentioned) $\mathcal{L}$ is non-Lipschitz and hence $\mathcal{L} \circ \tilde{\mathcal{F}}$ can “look” very different from $\tilde{\mathcal{F}}$. To deal with this, we approximate our 0-1 loss $\mathcal{L}$ by a Lipschitz “margin loss” function φ_δ , defined by

$$\varphi_\delta(u) = \begin{cases} 1 & \text{if } u \leq 0, \\ 1 - u/\delta & \text{if } 0 < u \leq \delta, \\ 0 & \text{if } u > \delta. \end{cases}$$

Taking $u = Y_i f(X_i)$, we see that φ_δ penalizes a classifier f for incorrect classification of the sample X_i , partially penalizes it for correct classification with a low margin (“low confidence”), and rewards it for correct classification with a high margin (“high confidence”). By contrast, the 0-1 loss $\mathcal{L}$ only penalizes incorrect classification. Notice two key properties of φ_δ : it is an upper bound for $\mathcal{L}$ (i.e. $\mathbf{1}(u \leq 0) \leq \varphi_\delta(u)$ for all $u \in \mathbb{R}$), and it is Lipschitz with constant $1/\delta$ (i.e. $|\varphi_\delta(u) - \varphi_\delta(v)| \leq \frac{1}{\delta} |u - v|$ for every $u, v \in \mathbb{R}$). By the first property we have

$$\mathcal{E}(f) = \mathbb{E}[\mathbb{I}(Yf(X) \leq 0)] \leq \mathbb{E}[\varphi_\delta(Yf(X))].$$

Moreover, since the range of φ_δ is contained in $[0, 1]$, it satisfies the same contractive property as $\mathcal{L}$ (replacing one of the data points can change the empirical error by at most $1/n$) and hence we can apply the entire machinery developed thus far to the function class $\varphi_\delta \circ \mathcal{F}$ to bound

$$(6) \quad \mathcal{E}(f) \leq \frac{1}{n} \sum_{i=1}^n \varphi_\delta(Y_i f(X_i)) + \sqrt{2\pi} G_n(\varphi_\delta \circ \tilde{\mathcal{F}}) + \frac{t}{\sqrt{n}}$$

with high probability (at least $1 - e^{-2t^2}$) for every $f \in \mathcal{F}$. At first glance, this may seem undesirable since our training error is now different, but this is actually fine since many learning algorithms are empirically known to be “high-margin classifiers” in the sense that they naturally try to make $Yf(X)$

large and positive. In other words, we don't have to worry too much about the effect of the margin penalty imposed by φ_δ : in most practical applications, if the training error measured by the 0-1 loss $\mathcal{L}$ is small, then it will also be small when measured by the margin loss φ_δ . Thus, we haven't "lost" anything with this new bound. Now that we have a Lipschitz function in place of $\mathcal{L}$, however, we can apply the following result:

Lemma (Gaussian Comparison Inequality). *Suppose that $\varphi : \mathbb{R} \rightarrow [0, 1]$ is Lipschitz with constant $L(\varphi)$. Then*

$$G_n(\varphi \circ \mathcal{H}) \leq L(\varphi)G_n(\mathcal{H}).$$

Proof. The proof will be an application of Slepian's lemma. Recall that this says if $(U_i)_{i \in I}$ and $(V_i)_{i \in I}$ are centered Gaussian processes indexed by I with the property that

$$\mathbb{E}(U_i - U_j)^2 \leq \mathbb{E}(V_i - V_j)^2 \quad \text{for all } i, j \in I,$$

then $\mathbb{E}[\sup_{i \in I} U_i] \leq \mathbb{E}[\sup_{i \in I} V_i]$. With this in mind, let $g_1, \dots, g_n$ be i.i.d. standard Gaussians that are independent of the X_i and defined on a separate probability space $(\Omega_g, \Sigma_g, \mathbb{P}_g)$. For fixed realizations of the X_i , consider the centered Gaussian processes $(U(h))_{h \in \mathcal{H}}$ and $(V(h))_{h \in \mathcal{H}}$ defined by

$$U(h) := \frac{1}{n} \sum_{i=1}^n g_i \varphi(h(X_i)) \quad \text{and} \quad V(h) := \frac{L(\varphi)}{n} \sum_{i=1}^n g_i h(X_i).$$

Notice that for any $f, h \in \mathcal{H}$ we have that

$$\begin{aligned} \mathbb{E}_g(U(f) - U(h))^2 &= \frac{1}{n^2} \sum_{i=1}^n [\varphi(f(X_i)) - \varphi(h(X_i))]^2 \\ &\leq \frac{L(\varphi)^2}{n^2} \sum_{i=1}^n (f(X_i) - h(X_i))^2 = \mathbb{E}_g(V(f) - V(h))^2. \end{aligned}$$

By Slepian's lemma, we subsequently deduce that

$$\mathbb{E}_g \left[\sup_{h \in \mathcal{H}} \frac{1}{n} \sum_{i=1}^n g_i \varphi(h(X_i)) \right] = \mathbb{E}_g \left[\sup_{h \in \mathcal{H}} U(h) \right] \leq \mathbb{E}_g \left[\sup_{h \in \mathcal{H}} V(h) \right] = L(\varphi) \mathbb{E}_g \left[\sup_{h \in \mathcal{H}} \frac{1}{n} \sum_{i=1}^n g_i h(X_i) \right]$$

for every fixed realization of $X_1, \dots, X_n$. By taking the expectation $\mathbb{E}_X$ over the X_i , the lemma follows. $\square$

Remark. The fact that we've defined the generalization gap $Z = \sup_{f \in \mathcal{F}} (\mathcal{E}(f) - \mathcal{E}_n(f))$ as a *one-sided* supremum actually simplifies things here. If we had instead worked with $\sup_{f \in \mathcal{F}} |\mathcal{E}(f) - \mathcal{E}_n(f)|$, then our previous work would have resulted in a bound in terms of the two-sided Gaussian complexity $\mathbb{E} \sup_{f \in \mathcal{F}} \frac{1}{n} |\sum_{i=1}^n g_i f(X_i)|$. A similar comparison inequality holds for this quantity as well (and can also be proven via Slepian's lemma), but the absolute values result in bound that contains some extra terms that we don't have to worry about in the one-sided case.

By applying the above to (6), we obtain that for every $f \in \mathcal{F}$ and $\delta > 0$, the following holds with probability at least $1 - e^{-2t^2}$:

$$(7) \quad \mathcal{E}(f) \leq \frac{1}{n} \sum_{i=1}^n \varphi_\delta(Y_i f(X_i)) + \frac{\sqrt{2\pi}}{\delta} G_n(\tilde{\mathcal{F}}) + \frac{t}{\sqrt{n}}.$$

At this point, we note that we can actually replace $G_n(\tilde{\mathcal{F}})$ with $G_n(\mathcal{F})$ since the two are equal. Indeed, we have that

$$G_n(\tilde{\mathcal{F}}) = \mathbb{E}_{X,g} \left[\sup_{f \in \mathcal{F}} \frac{1}{n} \sum_{i=1}^n g_i Y_i f(X_i) \right] = \mathbb{E}_X \mathbb{E}_g \left[\sup_{f \in \mathcal{F}} \frac{1}{n} \sum_{i=1}^n g_i Y_i f(X_i) \right],$$

and for fixed $\{(X_i, Y_i)\}_{i=1}^n$, $\{g_i Y_i\}_{i=1}^n$ and $\{g_i\}_{i=1}^n$ have the same distribution.

Note that when n is large, (7) says that the generalization error is highly likely to be bounded by the empirical margin error (which, as previously mentioned, is made small via the training process) plus the Gaussian complexity of our base class $\mathcal{F}$. Thus, the problem of ensuring good generalization essentially boils down to bounding this Gaussian complexity term. If the class $\mathcal{F}$ is sufficiently simple that this can be made to be small, then it will follow that *every* $f \in \mathcal{F}$ (i.e. no matter what classifier is produced by our learning algorithm) will generalize well with high probability.

3. APPLICATION: NEURAL NETWORKS

In this section, we apply the general theory we’ve just developed to bound the generalization error of special class of neural networks. Specifically, we consider fully-connected, ℓ -layer neural networks with bounded weights, defined by the following ingredients:

- (1) **Hyperparameters:** Independent of the network, we fix a “base class” $\mathcal{H}$ of measurable functions $S \rightarrow \{-1, 1\}$ and positive constants $A_1, \dots, A_\ell > 0$.
- (2) **Architecture:** An acyclic, directed graph G with vertex set divided into layers as $V(G) = \bigcup_{j=0}^\ell V_j$. This consists of an *input layer* V_0 (the vertices of which receives the raw input $x \in S$), $\ell - 1$ *hidden layers* $V_1, \dots, V_{\ell-1}$, and an *output layer* $V_\ell = \{v_o\}$ consisting of a single vertex that produces the final prediction of the network. Each *neuron* (vertex) in a given layer is connected to *every* vertex in the next layer.
- (3) **Parameters:** Each neuron $v \in V_0$ of the input layer is labelled by a member $h \in \mathcal{H}$ of the base class. Moreover, each neuron v in the subsequent layers $V_1, \dots, V_\ell$ is labelled by a *weight vector* $w := (w_1, \dots, w_{|V_{j-1}|}) \in \mathbb{R}^{|V_{j-1}|}$, which we think of as measuring the strength of connection between v and the neurons of the previous layer. Finally, for each layer $j = 1, \dots, \ell$, we require that $\|w_v\|_{\ell^1} \leq A_j$ for every $v \in V_j$.
- (4) **Computation:** The network works recursively in terms of its architecture as follows:
 - Each neuron $v \in V_0$ of the input layer receives the input $x \in S$, computes it using the function $h \in \mathcal{H}$ it is labelled with and passes this value along its edges to the neurons of V_1 .
 - Each neuron $v \in V_j$ (the j th layer, for $j \geq 1$) computes and passes along its outgoing edges the value

$$N_{\sigma, w}(u_1, \dots, u_{|V_{j-1}|}) = \sigma \left(\sum_{j=1}^{|V_{j-1}|} w_j u_j \right) = \sigma(w \cdot u),$$

where $\sigma : x \mapsto \max\{x, 0\}$ is the “ReLU” function (or *activation* function), the u_j are the inputs v receives from the previous layer, and w is the weight it is labelled with.

- The final prediction $f(x)$ of the network is the value computed by the output neuron v_o .

Recall from Section 2 that our underlying function class $\mathcal{F}$ is supposed to represent the set of all possible classifiers that could be produced by our learning algorithm. In the case of neural networks as defined here, the “learned” function f is completely determined by the base class $\mathcal{H}$ and the weights associated to the neurons (which, in practice, are what is adjusted during the training process). In particular, if we recursively define $\mathcal{H}_0 := \mathcal{H}$ and

$$\mathcal{H}_j := \{N_{\sigma, w}(h_1, \dots, h_m) : m \geq 0, h_i \in \mathcal{H}_{j-1}, w \in \mathbb{R}^m, \|w\|_{\ell^1} \leq A_j\},$$

then the function class $\mathcal{H}_\ell$ is precisely the class we’re interested in: the set of all possible prediction functions f computable by a fully-connected, ℓ -layer neural network with base class $\mathcal{H}$ and weight bounds $A_1, \dots, A_\ell$.

By applying the machinery developed in Section 2, we know that for every $h \in \mathcal{H}_\ell$ the following holds with probability at least $1 - e^{-2t^2}$:

$$\mathcal{E}(h) \leq \frac{1}{n} \sum_{i=1}^n \varphi_\delta(Y_i h(X_i)) + \frac{\sqrt{2\pi}}{\delta} G_n(\mathcal{H}_\ell) + \frac{t}{\sqrt{n}}.$$

The goal of this section is to explicitly bound the complexity term $G_n(\mathcal{H}_\ell)$ in our particular case. The key observation that will allow us to do this is that even though in general the class $\mathcal{H}_\ell$ can be quite complex (neural networks, especially deep ones, are capable of capturing a wide variety of patterns), the non-linearity introduced by the activation function σ is Lipschitz and the weights in each layer have uniformly bounded ℓ^1 -norms, so the complexity cannot increase “too much” from layer to layer. Specifically, we have the following:

Lemma. *For each $j \geq 1$, $G_n(\mathcal{H}_j) \leq 2A_j G_n(\mathcal{H}_{j-1})$.*

Proof. For $j \geq 1$, define

$$\mathcal{H}'_j = \{N_w(h_1, \dots, h_m) : m \geq 0, h_i \in \mathcal{H}_{j-1}, w \in \mathbb{R}^m, \|w\|_{\ell^1} \leq A_j\},$$

where $N_w(u_1, \dots, u_m) = \sum_{j=1}^m w_j u_j$. Then clearly $\mathcal{H}_j = \sigma \circ \mathcal{H}'_j$, so by the Gaussian comparison inequality we have that $G_n(\mathcal{H}_j) \leq G_n(\mathcal{H}'_j)$ (recall that σ is 1-Lipschitz). The proof will therefore be complete if we can show that $G_n(\mathcal{H}'_j) \leq 2A_j G_n(\mathcal{H}_{j-1})$. Indeed,

$$\begin{aligned} G_n(\mathcal{H}'_j) &\leq \mathbb{E} \sup_{h \in \mathcal{H}'_j} \left| \frac{1}{n} \sum_{i=1}^n g_i h(X_i) \right| \\ &= \mathbb{E} \sup_{m \geq 0, w \in \mathbb{R}^m, h_k \in \mathcal{H}_{j-1}} \left| \frac{1}{n} \sum_{i=1}^n g_i \sum_{k=1}^m w_k h_k(X_i) \right| \\ &= \mathbb{E} \sup_{m \geq 0, w \in \mathbb{R}^m, h_k \in \mathcal{H}_{j-1}} \left| \sum_{k=1}^m w_k \frac{1}{n} \sum_{i=1}^n g_i h_k(X_i) \right|. \end{aligned}$$

Notice that for each fixed choice of $m \geq 0$, $w \in \mathbb{R}^m$, and $h_1, \dots, h_m \in \mathcal{H}_{j-1}$ we have (treating the g_i and X_i as fixed as well, which is valid since we are working inside an expectation)

$$\left| \sum_{k=1}^m w_k \frac{1}{n} \sum_{i=1}^n g_i h_k(X_i) \right| \leq \sum_{k=1}^m |w_k| \left| \frac{1}{n} \sum_{i=1}^n g_i h_k(X_i) \right| \leq A_j \sup_{h \in \mathcal{H}_{j-1}} \left| \frac{1}{n} \sum_{i=1}^n g_i h(X_i) \right|$$

Since this holds for *every* choice of $m \geq 0$, $w \in \mathbb{R}^m$, and $h_1, \dots, h_m \in \mathcal{H}_{j-1}$, we deduce that

$$\mathbb{E} \sup_{m \geq 0, w \in \mathbb{R}^m, h_k \in \mathcal{H}_{j-1}} \left| \sum_{k=1}^m w_k \frac{1}{n} \sum_{i=1}^n g_i h_k(X_i) \right| \leq A_j \mathbb{E} \sup_{h \in \mathcal{H}_{j-1}} \left| \frac{1}{n} \sum_{i=1}^n g_i h(X_i) \right|$$

and it remains to observe that (by symmetry of Gaussians)

$$\begin{aligned} \mathbb{E} \sup_{h \in \mathcal{H}_{j-1}} \left| \frac{1}{n} \sum_{i=1}^n g_i h(X_i) \right| &\leq \mathbb{E} \left[\sup_{h \in \mathcal{H}_{j-1}} \frac{1}{n} \sum_{i=1}^n g_i h(X_i) \right] + \mathbb{E} \left[\sup_{h \in -\mathcal{H}_{j-1}} \frac{1}{n} \sum_{i=1}^n g_i h(X_i) \right] \\ &= 2\mathbb{E} \left[\sup_{h \in \mathcal{H}_{j-1}} \frac{1}{n} \sum_{i=1}^n g_i h(X_i) \right] = 2G_n(\mathcal{H}_{j-1}). \end{aligned}$$

□

By iteratively applying the above, our bound becomes

$$\mathcal{E}(h) \leq \frac{1}{n} \sum_{i=1}^n \varphi_\delta(Y_i h(X_i)) + \frac{\sqrt{2\pi}}{\delta} \left(\prod_{j=1}^\ell A_j \right) G_n(\mathcal{H}_0) + \frac{t}{\sqrt{n}} \quad \text{with probability at least } 1 - e^{-2t^2},$$

valid for every $h \in \mathcal{H}_\ell$. We have thus reduced the problem of bounding the generalization error in this case to bounding the Gaussian complexity $G_n(\mathcal{H}_0)$ of the base class $\mathcal{H}_0$. Since this class is usually relatively simple, we will bound it in terms of a combinatorial notion of complexity called VC-dimension, defined as follows:

Definition. Let $\mathcal{C}$ be a class of sets. We say that $\mathcal{C}$ *shatters* the set X if

$$\mathcal{C} \cap X := \{C \cap X : C \in \mathcal{C}\} = \mathcal{P}(X).$$

The *VC-dimension* V of the class $\mathcal{C}$ is the cardinality of the largest set shattered by $\mathcal{C}$, that is,

$$V := \sup |X|$$

where the supremum is taken over the class of all sets shattered by $\mathcal{C}$.

The key result that will allow us to bound the Gaussian complexity in terms of the VC-dimension of the class $\mathcal{H}_0$ is the following:

Lemma (Sauer's Lemma). *If $\mathcal{C}$ is a collection of subsets of X with finite VC-dimension V , then for any $\{x_1, \dots, x_n\} \subset X$ we have that*

$$|\mathcal{C} \cap \{x_1, \dots, x_n\}| \leq \left(\frac{en}{V} \right)^V.$$

If $\mathcal{C}$ has VC-dimension V , then we know that it can shatter V points. The basic question that Sauer's lemma answers is: how many subsets can $\mathcal{C}$ pick out for $n > V$? If $\mathcal{C}$ had infinite VC-dimension (so that it can shatter sets of arbitrarily large cardinality), then the growth function

$$\Delta_n(\mathcal{C}) := \sup_{\{x_1, \dots, x_n\} \subset X} |\mathcal{C} \cap \{x_1, \dots, x_n\}|$$

is *exponential* in n (indeed, it is equal to 2^n for every n). The significance of Sauer's lemma is that if $\mathcal{C}$ is instead "simple" enough that it has *finite* VC-dimension, then $\Delta_n(\mathcal{C})$ can only grow *polynomially* in n , that is, it doesn't have the "complexity" to overfit too much. This ultimately will allow us to obtain a good bound on $G_n(\mathcal{H}_0)$ by combining classical Gaussian concentration results with a union bound.

To see this, recall that we wish to bound

$$G_n(\mathcal{H}_0) = \mathbb{E}_X \mathbb{E}_g \sup_{h \in \mathcal{H}_0} \left| \frac{1}{n} \sum_{i=1}^n g_i h(X_i) \right|.$$

Let us first bound the inner expectation for a fixed realization of X . Notice that even though the base class $\mathcal{H}_0$ may be infinite, for fixed X the set of possible patterns

$$\mathcal{H}_0(X_1, \dots, X_n) := \{(h(X_1), \dots, h(X_n)) : h \in \mathcal{H}_0\}$$

is finite because the $h \in \mathcal{H}_0$ are $\{\pm 1\}$ -valued. Note, moreover, that

$$|\mathcal{H}_0(X_1, \dots, X_n)| = |\mathcal{C} \cap \{X_1, \dots, X_n\}|$$

for $\mathcal{C} = \{h(x) = 1 : h \in \mathcal{H}_0\}$ (this is because a labelling $(h(X_1), \dots, h(X_n))$ is determined uniquely by which $h(X_i)$ equal 1). By Sauer's lemma, we thus have

$$|\mathcal{H}_0(X_1, \dots, X_n)| \leq \left(\frac{en}{V} \right)^V$$

for V given by the VC-dimension of $\mathcal{C}$. If we now decompose

$$\left\{ \sup_{h \in \mathcal{H}_0} \left| \frac{1}{n} \sum_{i=1}^n g_i h(X_i) \right| > t \right\} = \bigcup_{h \in \mathcal{H}_0(X_1, \dots, X_n)} \left\{ \left| \frac{1}{n} \sum_{i=1}^n g_i h(X_i) \right| > t \right\},$$

then by a union bound we obtain

$$\begin{aligned} \mathbb{P} \left(\sup_{h \in \mathcal{H}_0} \left| \frac{1}{n} \sum_{i=1}^n g_i h(X_i) \right| > t \right) &\leq \sum_{h \in \mathcal{H}_0(X_1, \dots, X_n)} \mathbb{P} \left(\left| \frac{1}{n} \sum_{i=1}^n g_i h(X_i) \right| > t \right) \\ &\leq |\mathcal{H}_0(X_1, \dots, X_n)| \sup_{h \in \mathcal{H}_0(X_1, \dots, X_n)} \mathbb{P} \left(\left| \frac{1}{n} \sum_{i=1}^n g_i h(X_i) \right| > t \right) \\ &\leq \left(\frac{en}{V} \right)^V \sup_{h \in \mathcal{H}_0(X_1, \dots, X_n)} \mathbb{P} \left(\left| \frac{1}{n} \sum_{i=1}^n g_i h(X_i) \right| > t \right). \end{aligned}$$

On the other hand, no matter the pattern $(h(X_1), \dots, h(X_n)) \in \mathcal{H}_0(X_1, \dots, X_n)$ we have that

$$\frac{1}{n} \sum_{i=1}^n g_i h(X_i) \sim \mathcal{N}(0, 1/n).$$

Accordingly, the standard tail bound for Gaussians yields

$$\sup_{h \in \mathcal{H}_0(X_1, \dots, X_n)} \mathbb{P} \left(\left| \frac{1}{n} \sum_{i=1}^n g_i h(X_i) \right| > t \right) = \sup_{h \in \mathcal{H}_0(X_1, \dots, X_n)} \mathbb{P}(|\mathcal{N}(0, 1/n)| > t) \leq 2e^{-nt^2/2}.$$

Putting everything together, we thus have

$$\mathbb{P} \left(\sup_{h \in \mathcal{H}_0} \left| \frac{1}{n} \sum_{i=1}^n g_i h(X_i) \right| > t \right) \leq 2 \left(\frac{en}{V} \right)^V e^{-nt^2/2}.$$

In fact, because probabilities are always bounded by 1, we can strengthen this bound slightly to $\min\{Ce^{-nt^2/2}, 1\}$ for $C = 2(en/V)^V$. With this, we deduce that

$$\mathbb{E}_g \sup_{h \in \mathcal{H}_0} \left| \frac{1}{n} \sum_{i=1}^n g_i h(X_i) \right| = \int_0^\infty \mathbb{P} \left(\sup_{h \in \mathcal{H}_0} \left| \frac{1}{n} \sum_{i=1}^n g_i h(X_i) \right| > t \right) dt \leq \int_0^\infty \min\{Ce^{-nt^2/2}, 1\} dt.$$

Notice that this bound is independent of X , so in fact it bounds the expectation $\mathbb{E}_X$ over X as well (i.e. $G_n(\mathcal{H}_0)$). Our final step in the analysis is thus to estimate the right-hand side. To this end, we note that

$$Ce^{-nt^2/2} \leq 1 \iff \frac{nt^2}{2} \geq \log C \iff t \geq \sqrt{\frac{2 \log C}{n}} := t_*$$

and so

$$G_n(\mathcal{H}_0) \leq \int_0^{t_*} 1 dt + \int_{t_*}^{\infty} Ce^{-nt^2/2} dt = \sqrt{\frac{2 \log C}{n}} + \frac{C}{\sqrt{n}} \int_{\sqrt{2 \log C}}^{\infty} e^{-t^2/2} dt \lesssim \sqrt{\frac{V \log(n/V)}{n}}.$$

With this estimate in place, our bound for the generalization error becomes

$$\mathcal{E}(h) \leq \frac{1}{n} \sum_{i=1}^n \varphi_{\delta}(Y_i h(X_i)) + \frac{C'}{\delta} \left(\prod_{j=1}^{\ell} A_j \right) \sqrt{\frac{V \log(n/V)}{n}} + \frac{t}{\sqrt{n}},$$

holding with probability at least $1 - e^{-2t^2}$ for every $h \in \mathcal{H}_{\ell}$.

We see from this analysis that in order to generalize well, a fully-connected, ℓ -layer neural network needs:

- Low training error with a confidence margin δ (achieved by high-margin training);
- Controlled weight norms;
- A simple base class $\mathcal{H}$ (in the sense of having low VC-dimension V); and,
- A large number n of training samples.

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